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Multilingual Hallucination Gaps in Large Language Models

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Abstract

Large language models (LLMs) are increasingly used as alternatives to traditional search engines given their capacity to generate text that resembles human language. However, this shift is concerning, as LLMs often generate hallucinations—misleading or false information that appears highly credible. In this study, we explore the phenomenon of hallucinations across multiple languages in free-form text generation, focusing on what we call *multilingual hallucination gaps*. These gaps reflect differences in the frequency of hallucinated answers depending on the prompt and language used. To quantify such hallucinations, we used the FACTSCORE metric and extended its framework to a multilingual setting. We conducted experiments using LLMs from the LLaMA, Qwen, and Aya families, generating biographies in 19 languages and comparing the results to Wikipedia pages. Our results reveal variations in hallucination rates, especially between high- and low-resource languages, raising important questions about LLM multilingual performance and the challenges in evaluating hallucinations in multilingual free-form text generation.

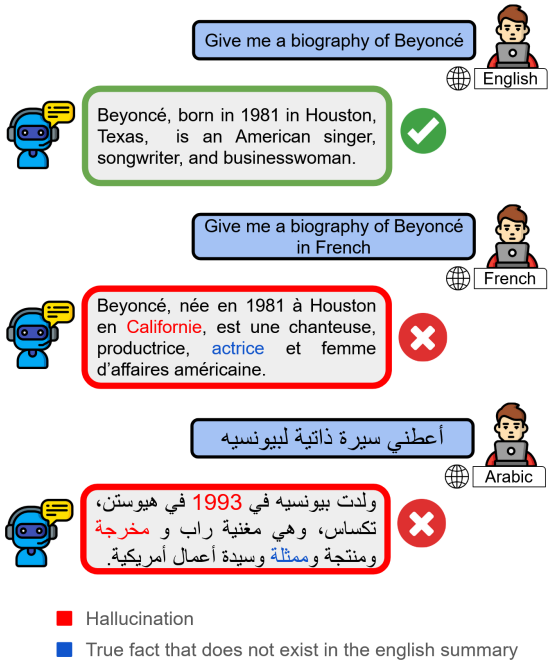

Figure 1: Example of Factual Hallucinations Gaps between languages

1 Introduction

Since the public release of ChatGPT, large language models (LLMs) have gained popularity. They are increasingly being integrated into or even replacing traditional search engines, such as the LLaMA model for Meta mobile applications or Gemma for Google. This trend shows an increasing reliance on LLMs as sources of knowledge, due to their ability to generate human-like text. However, such use is concerning as LLMs tend to produce hallucinations.

A hallucination occurs when a LLM generates *false* content (Rawte et al., 2023) with respect to a specific *reference*. Based on the reference type, hallucinations can be classified as follows (Zhang et al., 2023c): input-conflicting, where the generated content contradicts the user's input; context-conflicting, where it contradicts earlier outputs from the model; and fact-conflicting, where it contradicts established external knowledge. This work focuses exclusively on fact-conflicting hallucinations.

Understanding and tackling the issue of hallucinations in LLMs bring unique challenges. For example, detecting hallucinations is inherently difficult as they often appear highly credible. The wide range of tasks that LLMs are applied to also adds to the complexity, making it harder to comprehensively evaluate and mitigate hallucinations across different applications (Zhang et al., 2023c).

Besides these well-known and investigated issues, hallucinations are also not produced in the same way depending on the prompt fed to the model. We introduce the concept of *multilingual hallucination gaps*, which refers to variations in the proportion of hallucinated outputs generated in response to prompts in different languages.

Measuring these gaps can reveal that prompts in certain languages are more likely to induce hallucinations than others, which can significantly impact the reliability and trustworthiness of LLMs, especially in low-resource languages.

Previous work (Hong et al., 2024; Lin et al., 2022) focused on measuring hallucinations through benchmarks that require human annotations, which can be costly and hard to scale for multilingual LLMs. These benchmarks are also not suited for a free-form text generation setup. As a result, an automated evaluation pipeline becomes highly desirable. Statistical measures like ROUGE fail to capture semantic variations (Sellam et al., 2020) while NLI-based approaches transfer poorly to these tasks (Falke et al., 2019). Among LLM-based methods, Chen et al., 2024 proposed an eigenvalue metric to measure self-consistent hallucination. However, measuring this metric is costly and might not be suitable for the evaluation of free-form generated text across multiple languages.

Consequently, we explore FACTSCORE (Min et al., 2023), a different LLM-based method to evaluate hallucinations. In particular, the FACTSCORE metric uses an LLM to fact-check outputs of other LLMs against a knowledge source. As the FACTSCORE was only developed and tested on English text, we extend the methodology to encompass different languages by comparing toknowledge sources in various languages and leveraging translation.

We evaluate LLMs from the LLaMA, Qwen and Aya families. We prompt them to generate biographies in 19 different languages and we then compute the FACTSCORE metric for each answer by comparing it to an external knowledge source, Wikipedia for this project. The computation is done through three different experimental setups. We finally analyzed the results with respect to the target language, to the experimental setup and to the LLM used for text generation. Our results show gaps in the FACTSCORE metric distribution across the prompt languages, particularly between high, medium, and low-resource languages.
Our main contributions are:

1. Extending the FACTSCORE framework to a multilingual setting to quantify hallucination gaps across languages, with a focus on the disparities between high-resource and low-resource languages;

2. Evaluating a range of open-source and multilingual models to investigate improvements associated with different architectures and model sizes;

3. Assessing the robustness of the FACTSCORE framework across knowledge sources, prompt languages, and prompt templates.

2 Related work

Evaluating hallucinations

Previous research has concentrated on evaluating, explaining, and mitigating hallucinations in language models (Ji et al., 2023; Zhang et al., 2023c). All these efforts have been focused on detecting hallucinations in English-generated text exclusively.

There are several human-annotated benchmarks available for this purpose, including those compiled in the unified benchmark on HuggingFace by Hong et al. (2024). Since these benchmarks rely on human annotation, they usually focus on short answers and are time-consuming to create, making them ill-suited for evaluating multilingual hallucinations in free-form text generation.

Automatic metrics for measuring hallucinations encompass statistical and model-based ones (Ji et al., 2023), many of which draw inspiration from summarization evaluation. Statistical metrics like ROUGE can only handle lexical information and fail to deal with syntactic or semantic variations (Sellam et al., 2020). NLI-based approaches are robust to lexical variability, but NLI models transfer poorly to abstractive summarization (Falke et al., 2019) and struggle to locate specific errors in generated content. Faithfulness Classification metrics (Liu et al., 2022) address this issue, but they rely heavily on English-annotated datasets.

Among LLM-based methods, Chen et al. (2024) proposed an eigenvalue-based metric for detecting self-consistent hallucinations. However, this approach is not well-suited for free-form text generation, where repeated prompts can produce different, yet correct, responses, leading to lower scores despite valid outputs. More relevant to long-form text generation are the methods proposed by Min et al. (2023) and Farquhar et al. (2024), both of which

decompose answers into atomic facts. Min et al. (2023) employs a LLM to fact-check these facts against a knowledge source, while Farquhar et al. (2024) uses semantic entropy probabilities. In this study, we adopt the FACTSCORE (Min et al., 2023) approach, computationally less expensive.

Multilingual LLM

Several studies have focused on evaluating language generation models within a multilingual framework. Some of these datasets include M3Exam (Zhang et al., 2023a) for performance on human exam and Flores-101 (Goyal et al., 2022) for translation abilities. For the performance of the LLMs we used on these datasets, refer to Annex A. We can note that these evaluation metrics are still not consistently disclosed in technical reports or widely-recognized benchmarks. Despite covering a broad range of applications, these datasets do not cover hallucinations. However, they do provide evidence that LLMs exhibit different performance across different languages, which serves as motivation for our work.

An additional open research question concerns how multilingual abilities in these models are acquired (Zhang et al., 2023b; Wendler et al., 2024), as some models demonstrate proficiency in languages that are not officially supported. This observation motivated our decision to test models across a wide range of languages, even those not explicitly supported.

Hallucination metrics for multilingual generation

Kang et al. (2024) examine automatic hallucination detection metrics across different languages, including ROUGE, Named Entity Overlap, and the NLI-based SUMMAC score. Their findings show that these metrics do not correlate. Previous studies have suggested that these metrics may not be reliable for assessing hallucinations (Ji et al., 2023), which motivates our investigation into LLM-based metrics, specifically the FACTSCORE metric, to evaluate hallucinations across languages on a range of open-source models.

The most related work to ours is Shafayat et al. (2024), as the authors also study how to extend the FACTSCORE metric to a multilingual context. However, their methodology revolves around prompting in the original language and then translating generated content before assessing factuality. We broaden our investigation by adding an experiment that prompts in English while requesting answers in another language, as well as another experiment that directly compares generations to the original language Wikipedia page. Further, we investigate the reliability of this choice of knowledge source. We also explore a wider range of languages, a different set of entities beyond politicians, as well as multilingual open-source models instead of ChatGPT. Additionally, our work critically examines the robustness of the metric itself and identifies areas for improving its reliability.

3 Measuring factuality

To evaluate factual hallucinations in multilingual free-form text generation, we use the FACTSCORE metric (Min et al., 2023). This metric is particularly suited for our goal because it offers an intuitive, automated evaluation pipeline that can be easily adapted to different languages. By breaking down responses into atomic facts, FACTSCORE not only provides a more precise measure of factuality but also provides two key pieces of information: the factuality rate and the number of facts in the response.

Let's suppose we have a response $\mathcal{R}$ generated by an LLM, hereinafter referred to as LM_{SUBJ}. The FACTSCORE metric for this response $\mathcal{R}$ then consists of the following steps:

1. Decompose $\mathcal{R}$ into a set of atomic facts $\mathcal{A}(\mathcal{R})$. An atomic fact is a short sentence conveying a single piece of information. This is achieved by prompting an LLM, hereinafter referred to as LM_{EVAL}, to "Please breakdown the following sentence into independent facts" after showing it some decomposition examples.

2. Compare each fact $a \in \mathcal{A}(\mathcal{R})$ with an external knowledge source $\mathcal{C}$. To do so, we retrieve the proper passage from the source $\mathcal{C}$. We then construct a prompt by concatenating the retrieved passage, the given atomic fact and "True or False?". We then feed this prompt to an LLM. We use the same LLM that was used to decompose atomic facts, the LM_{EVAL}. The answer gives us:

$$\text{Supported}(a, \mathcal{C}) = \mathbb{1}\{a \text{ supported by } \mathcal{C}\}$$

3. (Optional) Add a length penalty p depending on a hyperparameter γ if our response $\mathcal{R}$ does